

MATH 3330 Homework 2

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I don't know why, but for whatever reason in my submission for Homework 1, my callout blocks for questions just displayed as empty (I write up these homework assignments in my note taking app Obsidian which works in markdown with LaTeX snippets). I think that this just might be a bug with how Gradescope renders the submitted PDFs since if you download it, you can see the question blocks perfectly clear. Anyways, I've changed a few settings from last time so they should hopefully appear now. 😊

Question 1

If f and g are surjective, what can you say about $g \circ f$? If $g \circ f$ is surjective, what can you say about f and g ? State and prove a theorem about surjectivity under composition.

If f and g are surjective, then $g \circ f$ must also be surjective.

Proof: Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions such that f and g are surjective. Take any $c \in C$. Since g is surjective, there exists some $b \in B$ such that $g(b) = c$ for all $c \in C$. Furthermore, since f is surjective, there exists some $a \in A$ such that $f(a) = b$ for all $b \in B$. Therefore the composition is $(g \circ f)(a) = g(f(a)) = g(b) = c$. Since $(g \circ f)(a) = c$ for all $c \in C$, then $g \circ f$ must be surjective.

If $g \circ f$ is surjective, then g must be surjective, but not necessarily f .

Proof: Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions such that $g \circ f$ is surjective. Let $f(a) = b$ for $a \in A$ and $b \in B$. For every $c \in C$, there exists some $a \in A$ such that $(g \circ f)(a) = g(f(a)) = g(b) = c$. Since $g(b) = c$ holds for all $c \in C$, g must be surjective.

Now consider f . Suppose for the sake of contradiction that f must be surjective.

Consider the example when $A = \{0\}$, $B = \{x, y\}$, and $C = \{\alpha\}$ and $f(0) = x$, $g(x) = \alpha$, and $g(y) = \alpha$. Then there exists a $b \in B$ (namely y) that f does not map to, meaning that f is not surjective.

Question 2

Show that f^{-1} preserves difference of sets, but not f .

Proof: We must show that $f^{-1}(A - B) = f^{-1}(A) - f^{-1}(B)$. Let $x \in f^{-1}(A - B)$. That means that $x \in A - B$. Thus, $x \in A$ and $x \notin B$. Since $x \in A$, $x \in f^{-1}(A)$, and since $x \notin B$, $x \notin f^{-1}(B)$. So, $x \in f^{-1}(A) - f^{-1}(B)$. Therefore, f^{-1} preserves difference of sets.

Now we must show that $f(A - B) \neq f(A) - f(B)$. Suppose for example $A = \{1, 2\}$ and $B = \{1\}$ such that $A, B \subseteq X$. We define $f : X \rightarrow Y$ by $f(1) = 0$ and $f(2) = 0$. On the left hand side, $f(A - B) = f(\{1, 2\} - \{1\}) = f(\{2\}) = \{0\}$. On the right hand side, $f(A) - f(B) = f(\{1, 2\}) - f(\{1\}) = \{0\} - \{0\} = \emptyset$. Since the left and right side are not equivalent, f does not preserve difference of sets.

Question 3

Let C be a relation on a set A . If $A_0 \subset A$, define the restriction of C to A_0 to be the relation $C \cap (A_0 \times A_0)$.

1. Show that the restriction of an equivalence relation is an equivalence relation.
2. Show that the restriction of an order relation is an order relation.

Proof: In order to show that the restriction of C to A_0 (we will define as $C_0 := C \cap (A_0 \times A_0)$) is an equivalence relation, we must show that it is reflexive, symmetric, and transitive. $(x, y) \in C_0$ means that $(x, y) \in C$, and that $x, y \in A_0$.

Reflexive: Let $x \in A_0$. Since $A_0 \subset A$, $x \in A$. Since C is an equivalence relation, it is reflexive on A , meaning $(x, x) \in C$. But also, $(x, x) \in (A_0 \times A_0)$, so $(x, x) \in C \cap (A_0 \times A_0) = C_0$. Therefore, the restriction of C to A_0 is reflexive.

Symmetric: Let $x, y \in A_0$ and suppose that $(x, y) \in C_0$. Then $(x, y) \in C$. Since C is an equivalence relation, it is symmetric on A , meaning $(y, x) \in C$. Because $x, y \in A_0$, $(y, x) \in (A_0 \times A_0)$, so $(y, x) \in C \cap (A_0 \times A_0) = C_0$. Therefore, the restriction of C to A_0 is symmetric.

Transitive: Let $x, y, z \in A_0$ and suppose that $(x, y) \in C_0$ and $(y, z) \in C_0$. Then $(x, y) \in C$ and $(y, z) \in C$. Since C is an equivalence relation, it is transitive on A , meaning $(x, z) \in C$. Because $x, z \in A_0$, $(x, z) \in (A_0 \times A_0)$, so $(x, z) \in C \cap (A_0 \times A_0) = C_0$. Therefore the restriction of C to A_0 is transitive.

Since the restriction of C to A_0 is reflexive, symmetric, and transitive, it is an equivalence relation.

Proof: In order to show that the restriction of C to A_0 is an order relation, we must show that it is reflexive, non-symmetric, and transitive. $(x, y) \in C_0 = C \cap (A_0 \times A_0)$ means that $(x, y) \in C$, and that $x, y \in A_0$.

Reflexive: Let $x \in A_0$. Since $A_0 \subset A$, $x \in A$. Since C is an order relation, it is reflexive on A , meaning $(x, x) \in C$. But also, $(x, x) \in (A_0 \times A_0)$, so $(x, x) \in C \cap (A_0 \times A_0) = C_0$. Therefore, the restriction of C to A_0 is reflexive.

Symmetric: Let $x, y \in A_0$ and suppose that $(x, y) \in C_0$ and $(y, x) \in C_0$. Then $(x, y) \in C$ and $(y, x) \in C$. Since C is an order relation and is non-symmetric on A , this implies that $x = y$, so therefore, the restriction of C to A_0 is also non-symmetric.

Transitive: Let $x, y, z \in A_0$ and suppose that $(x, y) \in C_0$ and $(y, z) \in C_0$. Then $(x, y) \in C$ and $(y, z) \in C$. Since C is an order relation, it is transitive on A , meaning $(x, z) \in C$. Because $x, z \in A_0$, $(x, z) \in (A_0 \times A_0)$, so $(x, z) \in C \cap (A_0 \times A_0) = C_0$. Therefore the restriction of C to A_0 is transitive.

Since the restriction of C to A_0 is reflexive, non-symmetric, and transitive, it is an order relation.

Question 4

Let S, S' be the following subsets of the plane:

$$S = \{(x, y) \mid y = x + 1 \text{ and } 0 < x < 2\}$$

$$S' = \{(x, y) \mid y - x \in \mathbb{Z}\}$$

1. Show that S' is an equivalence relation on the real line and that $S \subset S'$. Describe the equivalence classes of S' .
2. Show that given any collection of equivalence relations on a set A , their intersection is an equivalence relation on set A .
3. Describe the equivalence relation T on the real line that is the intersection of all equivalence relations on the real line that contain S . Describe the equivalence classes of T .

1.

We must show that S' is reflexive, symmetric, and transitive. For all x , $(x, x) = x - x = 0 \in S'$, so S' is reflexive. For all (x, y) , $(x, y) = y - x = -(x - y) = (-y, -x)$. Since $\mathbb{Z}$ is closed under multiplication by -1 , $(y, x) \in S'$, so S' is symmetric. For all (x, y) and (y, z) , we have $y - x \in \mathbb{Z}$ and $z - y \in \mathbb{Z}$. Then, $(y - x) + (z - y) = z - x$, so $x \sim z$ and thus S' is transitive. Therefore, S' is an equivalence relation.

Furthermore, if you pick some $(x, y) \in S$, by definition, $y = x + 1$ and $0 < x < 2$. Then, $y - x = 1 \in \mathbb{Z}$, so $(x, y) \in S'$. Therefore, $S \subset S'$.

The equivalence classes of S' form sets of numbers that differ by whole integer values (i.e. $\{1.5, 2.5, 8.5, \dots\}$).

2.

Let $\{\sim_i\}_{i \in I}$ be a collection of equivalence relations on a set A . Define a new relation $\sim$ by $x \sim y$ if and only if $x \sim_i y$ for every $i \in I$. This relation is exactly the intersection $\sim = \bigcap_{i \in I} \sim_i$.

For any $x \in A$, since each $\sim_i$ is an equivalence relation, we have $x \sim_i x$ for all $i \in I$. Therefore $x \sim x$, so $\sim$ is reflexive.

Suppose $x \sim y$. Then $x \sim_i y$ for every $i \in I$. Since each $\sim_i$ is symmetric, it follows that $y \sim_i x$ for every i , and thus $y \sim x$. Thus $\sim$ is symmetric.

Suppose $x \sim y$ and $y \sim z$. Then for every $i \in I$, we have $x \sim_i y$ and $y \sim_i z$. Because each $\sim_i$ is transitive, this implies $x \sim_i z$ for all i , and therefore $x \sim z$. Hence $\sim$ is transitive.

Since $\sim$ is reflexive, symmetric, and transitive, the intersection of any collection of equivalence relations on A is itself an equivalence relation.

3.

Let T be the intersection of all equivalence relations on $\mathbb{R}$ that contain S . T is itself an equivalence relation (from part 2), and by construction it is the smallest equivalence relation on $\mathbb{R}$ that contains S .

Since $S \subset S'$, and S' is an equivalence relation, $T \subset S'$. On the other hand, any equivalence relation that contains S must also contain all pairs from S by symmetry and transitivity. Because $(x, x+1) \in S$ for $0 < x < 2$, transitivity forces $(x, x+n)$ to be in the relation for all integers n , and symmetry gives $(x+n, x)$. Thus any equivalence relation containing S must contain all pairs (x, y) with $y - x \in \mathbb{Z}$.

Therefore $S' \subset T$, and hence $T = S'$.

The equivalence classes of T are sets of real numbers that differ by exactly an integer.

Question 5

Prove the following statement: If an ordered set A has the least upper bound property, then it has the greatest lower bound property.

Proof: Assume A is an ordered set with the least upper bound property. We must show that A also has the greatest lower bound property.

Let $B \subset A$ be a nonempty set that is bounded below. Let L be the set of all lower bounds of B . Since B is bounded below, L is nonempty. Pick any $b \in B$. Every element of L is less than or equal to b , so b is an upper bound of L , and thus L is bounded above. By the least upper bound property, L has a least upper bound. We will define this least upper bound as u .

We claim that u is the greatest lower bound of B . First, u is a lower bound of B . Since every element of L is less than or equal to every $b \in B$, each b is an upper bound of L , and because u is the least upper bound of L , we have $u \leq b$ for all $b \in B$. Next, u is the greatest such lower bound. If m is any lower bound of B , then $m \in L$, and since u is an upper bound of L , we have $m \leq u$.

Therefore u is the greatest lower bound of B , and A has the greatest lower bound property.

Question 6

Show that if $\mathcal{A}$ is a collection of inductive sets, then the intersection of the elements of $\mathcal{A}$ is also an inductive set (and thus $\mathbb{Z}_+$ is an inductive set).

Proof: A set S is inductive if $1 \in S$ and whenever $n \in S$, then $n + 1 \in S$. Let $\mathcal{A}$ be a collection of inductive sets and define

$$I = \bigcap_{A \in \mathcal{A}} A$$

as the intersection of the sets in $\mathcal{A}$. We must show that I is also inductive. Since each $A \in \mathcal{A}$ is an inductive set, $1 \in A$ for all $A \in \mathcal{A}$. Thus $1 \in I$.

Now, assume that $n \in I$, meaning that $n \in A$ for all $A \in \mathcal{A}$. But since every A is inductive, it's also necessary that $n + 1 \in A$ for all $A \in \mathcal{A}$. Since $n + 1 \in A$ for all $A \in \mathcal{A}$, then $n + 1 \in I$.

Therefore, I , the intersection of the elements of $\mathcal{A}$ is also an inductive set, and thus $\mathbb{Z}_+$.

Question 7

Prove by induction that given $n \in \mathbb{Z}_+$, every nonempty subset of $\{1, \dots, n\}$ has a largest element. Does this mean that every nonempty subset of $\mathbb{Z}_+$ has a largest element? Why or why not?

Proof: Let $n \in \mathbb{Z}_+$. We must show that every nonempty subset of $\{1, \dots, n\}$ has a largest element. Take $n = 1$. The set $\{1, \dots, n\}$ has exactly one nonempty subset, being $\{1\}$, which trivially has the largest element of 1. This completes the base case.

Now take $k = n + 1$ and assume that every subset of $\{1, \dots, n\}$ has a largest element. We must show that every nonempty subset of $\{1, \dots, n, k\}$ has a largest element. Every subset of $\{1, \dots, n, k\}$ must either contain k or not, so we have two cases:

1. $k \in S \subseteq \{1, \dots, n, k\}$, so k is the largest element, since $k = n + 1$ is larger than all elements in $S \subseteq \{1, \dots, n\}$.
2. $k \notin S \subseteq \{1, \dots, n, k\}$, so $S \subseteq \{1, \dots, n\}$. By our initial assumption, every subset of S has a largest element.

Since we have shown that every nonempty subset of $\{1, \dots, n\}$ has a largest element for $n = 1$ and have proven that this statement holds for $k = n + 1$, it must be true.

No, not every nonempty subset of $\mathbb{Z}_+$ has a largest element because if you take $\mathbb{Z}_+$ itself, there is no largest element since $\mathbb{Z}_+$ is an infinite set.

Question 8

Which of the following subsets of $\mathbb{R}^\omega$ can be written as the cartesian product of subsets of $\mathbb{R}$?

1. $\{x \mid x_i \text{ is an integer for all } i\}$
2. $\{x \mid x_2 = x_3\}$

1. **Yes:** If you take $A_i \in \mathbb{Z}$ for every i , then $\{x \mid x_i \text{ is an integer for all } i\} = \prod_{i \in \omega} \mathbb{Z} = \mathbb{Z}^\omega$. This works since $\mathbb{Z} \subset \mathbb{R}$.
2. **No:** This set forces two elements to be identical which can not be expressed in a cartesian product.

Question 9

Show that if B is not finite and $B \subset A$, then A must not be finite.

Proof: Let B be a non-finite set and $B \subset A$. Assume for the sake of contradiction that A is finite. Since A is finite, every subset of A must also be finite. However, $B \subset A$ and B is not finite, contradicting our claim that A is finite. Therefore, A must not be finite.